3.16 Principal Ideal Domains

Definition 3.16.1 (Principal Ideal Domain)
*An integral domain A is a **principal ideal domain** (or **PID**) if every ideal $\mathfrak{a}$ is principal.*

Proposition 3.16.2
A PID is Noetherian.

Proof. Suppose we have an ascending chain of ideals

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots \subseteq \mathfrak{a}_n \subseteq \cdots$$

Clearly the union is again an ideal, which is also principal of the form (a). We must have $a \in \mathfrak{a}_n$ for some n, whence it terminates after n. ∎

Proposition 3.16.3 (Integers form a PID)
$\mathbb{Z}$ is a PID.

Proof. This follows from the well-ordering principle. Let $\mathfrak{a}$ be an ideal with minimal positive element d. We claim $\mathfrak{a} = (d)$. By the division algorithm (or apply well-ordering principle to the coset $x + (d)$), for every $x \in \mathfrak{a}$ there is $0 \leq r < d$ and $q \in \mathbb{Z}$ such that

$$x = qd + r.$$

Clearly $r \in \mathfrak{a}$, whence by minimality $r = 0$ as required. ∎

Proposition 3.16.4
Let k be a field then the polynomial ring $k[X]$ is a PID

Proof. Let $\mathfrak{a} \triangleleft k[X]$ be an ideal. Choose $f \in \mathfrak{a}$ to have minimal degree, then we claim $\mathfrak{a} = (f)$. For $g \in \mathfrak{a}$ we have by (3.8.13) $g = qf + r$ for $\deg(r) < \deg(f)$. Clearly $r \in \mathfrak{a}$, so by minimality $r = 0$ and the result follows. ∎

Lemma 3.16.5 (Co-prime elements in a PID)
Let A be a PID, then x, y are coprime if and only if they have no non-invertible common divisors.

Proof. First suppose $(x, y) = A$, then $ax + by = 1$ and any common divisor d must divide 1 and therefore be invertible.

Conversely suppose $(x, y) \neq (1)$, since A is a PID it must equal (d) for some non-invertible d which is then a common divisor. ∎

Proposition 3.16.6
Let A be a principal ideal domain, M a free A-module with rank m and $N \subset M$ an A-submodule. Then N is free with rank $n \leq m$.

Proof. Recall (...) that the rank of a finite free A-module is well-defined.

We proceed by induction on m. Suppose that $m = 1$ then there is an isomorphism of A-modules $A \xrightarrow{\sim} M$. Therefore under this isomorphism N corresponds to an ideal, say (a). If $a = 0$ then $N = 0$ is by definition free of rank 0. If $a \neq 0$ then we may consider the map of A-modules

$$\begin{aligned} A &\to (a) \\ x &\to ax \end{aligned}$$

it is by definition surjective, and injective by cancellation and so an isomorphism. This shows that N is free of rank 1.

Suppose then $x_1, ..., x_m$ is a basis for M, and let M' be the free submodule generated by $x_1, ..., x_{m-1}$, and M'' the rank-1 free module generated by x_m. Then

$$M \coloneqq M' \oplus M''$$

and we may show that N is the internal direct sum

$$N = (N \cap M') \oplus (N \cap M'').$$

Then by the induction hypothesis we find N is free of rank at most $m - 1 + 1 = m$. ∎

Proposition 3.16.7
Let A be a principal ideal domain and M a finitely generated, torsion-free A-module. Then M is a finite free A-module.

Proof. Let $\{m_1, ..., m_n\}$ be a generating set for M, and $\{v_1, ..., v_r\}$ an arbitrary linearly independent subset of M. Then by definition we have

$$v_i = \sum_{j=1}^{n} a_{ij} m_j \quad a_{ij} \in A$$

Consider the elements of A^n given by

$$w_i := (a_{i1}, ..., a_{in})$$

Then we can see that $\{w_1, ..., w_r\}$ is an A-linearly independent subset of A^n, and therefore a K-linearly independent subset of K^n for $K = \mathrm{Frac}(A)$. By (...) we see that $r \leq n$.

Therefore we may choose $\{v_1, ..., v_r\}$ to be a maximal linearly independent subset and define $M' := \langle v_1, ..., v_r \rangle$. For each generator m_i, by maximality of the subset there must exist an $a_i \neq 0$ such that

$$a_i m_i \in M'$$

Define $a := a_1 ... a_n$ then this satisfies $aM \subseteq M'$, since the m_i generate M. As A is an integral domain we have a series of injections

$$M \hookrightarrow aM \subseteq M' \xrightarrow{\sim} A^r$$

Therefore we are done by (3.16.6). ∎